.NET Client Practicals

All 7 Web Service Methods — C# Console Application

Addition · Subtraction · Multiplication · Division · Factorial · Calculator · Prime Number

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Practical 1 — .NET Client

Calling Addition Web Service from C# Console App

Aim

To create a .NET Console Application that connects to the Addition web service and displays the result.

Steps

Step 1 — Open Visual Studio

1. Start Visual Studio
2. Click Create a new project

Step 2 — Create Console Application

3. In the search box type: console
4. Select: Console App (.NET Framework)
5. Click Next

Step 3 — Enter Project Details

- Project Name: AdditionClient
- Click: Create

Visual Studio will create the project.

Step 4 — Add Web Service Reference

In Solution Explorer (right side):

6. Right click the project
7. Click Add → Service Reference

A new window opens. In the Address box paste your WSDL URL:

```
http://localhost:8080/CurrencyService/_currency?wsdl
```

8. Click: Go — Visual Studio will detect the service

- Namespace: CurrencyRef

9. Click: OK

Now a folder Service References will appear in your project.

Step 5 — Write C# Code

Open: Program.cs

Replace the code with:

```
using System;
using AdditionClient.CurrencyRef;

namespace AdditionClient
{
    class Program
    {
        static void Main(string[] args)
        {
            CurrencyClient client = new CurrencyClient();
            Console.Write("Enter value of A: ");
            double a = Convert.ToDouble(Console.ReadLine());
            Console.Write("Enter value of B: ");
            double b = Convert.ToDouble(Console.ReadLine());
            double result = client.addition(a, b);
            Console.WriteLine("Addition Result = " + result);
            Console.ReadLine();
        }
    }
}
```

Step 6 — Run the Project

10. Press F5 or click the green Start button
11. A black console window will open
12. Enter the required values when prompted
13. Result will be displayed in the console

Expected Output

```
Enter value of A: 5
Enter value of B: 3
Addition Result = 8
```

Practical 2 — .NET Client

Calling Subtraction Web Service from C# Console App

Aim

To create a .NET Console Application that connects to the Subtraction web service and displays the result.

Steps

Step 1 — Open Visual Studio

14. Start Visual Studio
15. Click Create a new project

Step 2 — Create Console Application

16. In the search box type: console
17. Select: Console App (.NET Framework)
18. Click Next

Step 3 — Enter Project Details

- Project Name: SubtractionClient
- Click: Create

Visual Studio will create the project.

Step 4 — Add Web Service Reference

In Solution Explorer (right side):

19. Right click the project
20. Click Add → Service Reference

A new window opens. In the Address box paste your WSDL URL:

```
http://localhost:8080/CurrencyService/_currency?wsdl
```

21. Click: Go — Visual Studio will detect the service

- Namespace: CurrencyRef

22. Click: OK

Now a folder Service References will appear in your project.

Step 5 — Write C# Code

Open: Program.cs

Replace the code with:

```
using System;
using SubtractionClient.CurrencyRef;

namespace SubtractionClient
{
    class Program
    {
        static void Main(string[] args)
        {
            CurrencyClient client = new CurrencyClient();
```

```
            Console.Write("Enter value of A: ");
            double a = Convert.ToDouble(Console.ReadLine());
            Console.Write("Enter value of B: ");
            double b = Convert.ToDouble(Console.ReadLine());
            double result = client.subtraction(a, b);
            Console.WriteLine("Subtraction Result = " + result);
            Console.ReadLine();
        }
    }
}
```

Step 6 — Run the Project

23. Press F5 or click the green Start button
24. A black console window will open
25. Enter the required values when prompted
26. Result will be displayed in the console

Expected Output

```
Enter value of A: 10
Enter value of B: 4
Subtraction Result = 6
```

Practical 3 — .NET Client

Calling Multiplication Web Service from C# Console App

Aim

To create a .NET Console Application that connects to the Multiplication web service and displays the result.

Steps

Step 1 — Open Visual Studio

27. Start Visual Studio
28. Click Create a new project

Step 2 — Create Console Application

29. In the search box type: console
30. Select: Console App (.NET Framework)
31. Click Next

Step 3 — Enter Project Details

- Project Name: MultiplicationClient
- Click: Create

Visual Studio will create the project.

Step 4 — Add Web Service Reference

In Solution Explorer (right side):

32. Right click the project
33. Click Add → Service Reference

A new window opens. In the Address box paste your WSDL URL:

```
http://localhost:8080/CurrencyService/_currency?wsdl
```

34. Click: Go — Visual Studio will detect the service

- Namespace: CurrencyRef

35. Click: OK

Now a folder Service References will appear in your project.

Step 5 — Write C# Code

Open: Program.cs

Replace the code with:

```
using System;
using MultiplicationClient.CurrencyRef;

namespace MultiplicationClient
{
    class Program
    {
        static void Main(string[] args)
        {
            CurrencyClient client = new CurrencyClient();
            Console.Write("Enter value of A: ");
            double a = Convert.ToDouble(Console.ReadLine());
            Console.Write("Enter value of B: ");
            double b = Convert.ToDouble(Console.ReadLine());
            double result = client.multiplication(a, b);
            Console.WriteLine("Multiplication Result = " + result);
            Console.ReadLine();
        }
    }
}
```

Step 6 — Run the Project

36. Press F5 or click the green Start button
37. A black console window will open
38. Enter the required values when prompted

39. Result will be displayed in the console

Expected Output

```
Enter value of A: 5
Enter value of B: 6
Multiplication Result = 30
```

Practical 4 — .NET Client

Calling Division Web Service from C# Console App

Aim

To create a .NET Console Application that connects to the Division web service and displays the result.

Steps

Step 1 — Open Visual Studio

40. Start Visual Studio
41. Click Create a new project

Step 2 — Create Console Application

42. In the search box type: console
43. Select: Console App (.NET Framework)
44. Click Next

Step 3 — Enter Project Details

- Project Name: DivisionClient
- Click: Create

Visual Studio will create the project.

Step 4 — Add Web Service Reference

In Solution Explorer (right side):

45. Right click the project
46. Click Add → Service Reference

A new window opens. In the Address box paste your WSDL URL:

```
http://localhost:8080/CurrencyService/_currency?wsdl
```

47. Click: Go — Visual Studio will detect the service
- Namespace: CurrencyRef
48. Click: OK

Now a folder Service References will appear in your project.

Step 5 — Write C# Code

Open: Program.cs

Replace the code with:

```
using System;
using DivisionClient.CurrencyRef;

namespace DivisionClient
{
    class Program
    {
        static void Main(string[] args)
        {
            CurrencyClient client = new CurrencyClient();
            Console.Write("Enter value of A: ");
            double a = Convert.ToDouble(Console.ReadLine());
            Console.Write("Enter value of B: ");
            double b = Convert.ToDouble(Console.ReadLine());
            double result = client.division(a, b);
            Console.WriteLine("Division Result = " + result);
            Console.ReadLine();
        }
    }
}
```

Step 6 — Run the Project

49. Press F5 or click the green Start button
50. A black console window will open
51. Enter the required values when prompted
52. Result will be displayed in the console

Expected Output

```
Enter value of A: 10
Enter value of B: 2
Division Result = 5
```

Practical 5 — .NET Client

Calling Factorial Web Service from C# Console App

Aim

To create a .NET Console Application that connects to the Factorial web service and displays the result.

Steps

Step 1 — Open Visual Studio

53. Start Visual Studio
54. Click Create a new project

Step 2 — Create Console Application

55. In the search box type: console
56. Select: Console App (.NET Framework)
57. Click Next

Step 3 — Enter Project Details

- Project Name: FactorialClient
- Click: Create

Visual Studio will create the project.

Step 4 — Add Web Service Reference

In Solution Explorer (right side):

58. Right click the project
59. Click Add → Service Reference

A new window opens. In the Address box paste your WSDL URL:

```
http://localhost:8080/CurrencyService/_currency?wsdl
```

60. Click: Go — Visual Studio will detect the service

- Namespace: CurrencyRef

61. Click: OK

Now a folder Service References will appear in your project.

Step 5 — Write C# Code

Open: Program.cs

Replace the code with:

```
using System;
using FactorialClient.CurrencyRef;

namespace FactorialClient
```

```
{
    class Program
    {
        static void Main(string[] args)
        {
            CurrencyClient client = new CurrencyClient();
            Console.Write("Enter value of N: ");
            int n = Convert.ToInt32(Console.ReadLine());
            int result = client.factorial(n);
            Console.WriteLine("Factorial of " + n + " = " + result);
            Console.ReadLine();
        }
    }
}
```

Step 6 — Run the Project

62. Press F5 or click the green Start button
63. A black console window will open
64. Enter the required values when prompted
65. Result will be displayed in the console

Expected Output

```
Enter value of N: 5
Factorial of 5 = 120
```

Practical 6 — .NET Client

Calling Calculator Web Service from C# Console App

Aim

To create a .NET Console Application that connects to the Calculator web service and displays the result.

Steps

Step 1 — Open Visual Studio

66. Start Visual Studio
67. Click Create a new project

Step 2 — Create Console Application

68. In the search box type: console

69. Select: Console App (.NET Framework)
70. Click Next

Step 3 — Enter Project Details

- Project Name: CalculatorClient
- Click: Create

Visual Studio will create the project.

Step 4 — Add Web Service Reference

In Solution Explorer (right side):

71. Right click the project
72. Click Add → Service Reference

A new window opens. In the Address box paste your WSDL URL:

```
http://localhost:8080/CurrencyService/_currency?wsdl
```

73. Click: Go — Visual Studio will detect the service

- Namespace: CurrencyRef

74. Click: OK

Now a folder Service References will appear in your project.

Step 5 — Write C# Code

Open: Program.cs

Replace the code with:

```
using System;
using CalculatorClient.CurrencyRef;

namespace CalculatorClient
{
    class Program
    {
        static void Main(string[] args)
        {
            CurrencyClient client = new CurrencyClient();
            Console.Write("Enter value of A: ");
            double a = Convert.ToDouble(Console.ReadLine());
            Console.Write("Enter value of B: ");
            double b = Convert.ToDouble(Console.ReadLine());
            Console.Write("Enter operation (add/sub/mul/div): ");
            string op = Console.ReadLine();
            double result = client.calculator(a, b, op);
            Console.WriteLine("Calculator Result = " + result);
            Console.ReadLine();
        }
    }
}
```

Step 6 — Run the Project

75. Press F5 or click the green Start button
76. A black console window will open
77. Enter the required values when prompted
78. Result will be displayed in the console

Expected Output

```
Enter value of A: 10
Enter value of B: 5
Enter operation (add/sub/mul/div): add
Calculator Result = 15
```

Practical 7 — .NET Client

Calling PrimeNumber Web Service from C# Console App

Aim

To create a .NET Console Application that connects to the PrimeNumber web service and displays the result.

Steps

Step 1 — Open Visual Studio

79. Start Visual Studio
80. Click Create a new project

Step 2 — Create Console Application

81. In the search box type: console
82. Select: Console App (.NET Framework)
83. Click Next

Step 3 — Enter Project Details

- Project Name: PrimeNumberClient
- Click: Create

Visual Studio will create the project.

Step 4 — Add Web Service Reference

In Solution Explorer (right side):

84. Right click the project
85. Click Add → Service Reference

A new window opens. In the Address box paste your WSDL URL:

```
http://localhost:8080/CurrencyService/_currency?wsdl
```

86. Click: Go — Visual Studio will detect the service

- Namespace: CurrencyRef

87. Click: OK

Now a folder Service References will appear in your project.

Step 5 — Write C# Code

Open: Program.cs

Replace the code with:

```
using System;
using PrimeNumberClient.CurrencyRef;

namespace PrimeNumberClient
{
    class Program
    {
        static void Main(string[] args)
        {
            CurrencyClient client = new CurrencyClient();
            Console.Write("Enter value of N: ");
            int n = Convert.ToInt32(Console.ReadLine());
            string result = client.primeNumber(n);
            Console.WriteLine("Result: " + result);
            Console.ReadLine();
        }
    }
}
```

Step 6 — Run the Project

88. Press F5 or click the green Start button
89. A black console window will open
90. Enter the required values when prompted
91. Result will be displayed in the console

Expected Output

```
Enter value of N: 7
Result: Prime Number
```